Welded steel workbench

All steel: four square-tube legs, a welded top frame and lower frame, and a top and lower shelf of steel bars laid flat. Straight square cuts only: every rail butts onto a leg face.

✓ All 19 checks pass

Make it yours

Length 1500 mm · Depth 700 mm · Work height 900 mm · Rated load 300 kg · Gap between top bars 10 mm · Lower shelf yes

OVERALL
1500 × 700 × 900 mm
length × depth × height

MASS
95 kg
as built, with every part listed

PIECES
49 cut (5 different)
4 plates

WELDS
24 joints
about 3.8 m of weld

RATED LOAD ON THE TOP
factor 4.5
sags 1.7 mm

POINT LOAD AT A FRONT CORNER
factor 16.8
safety factor on yield

SHOVED SIDEWAYS AT THE TOP
factor 10.0
safety factor on yield

SHOVED FRONT TO BACK AT THE TOP
factor 11.8
safety factor on yield

LOADED LOWER SHELF
factor 19.4
sags 0.3 mm

BUILD TIME
20 h
9 steps

COST
~$239
priced lines only; see the materials list

WORK HEIGHT
900 mm
top surface, feet mid-travel

TOP
14 steel bars
RHS 40×20 laid flat, 10.8 mm gaps

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1 Checks

All 19 checks pass at this size.

	Check	Result
✓	**Every joint can be made as drawn** Each end is a mitre, a butt onto a face, a flush cut or a free end; nothing overhangs.	`24 joints`
✓	**Nothing floating** Every member, plate and part touches what holds it.	`all connected`
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	`none`
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	`longest 1500 mm`
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	`shortest 620 mm`
✓	**Rated load on the top: strength** Yield ÷ peak stress with 337 kg spread over the top frame, with the bars, plus self weight.	`safety factor 4.46 (need 2), worst A (leg)`
✓	**Rated load on the top: deflection** Largest movement of the loaded members.	`1.7 mm (limit 4.9 mm)`
✓	**Point load at a front corner: strength** Yield ÷ peak stress with 150 kg on one front corner, plus self weight.	`safety factor 16.78 (need 2), worst A (leg)`
✓	**Shoved sideways at the top: strength** Yield ÷ peak stress with 400 N pushing the top end-on, plus self weight.	`safety factor 9.95 (need 2), worst A (leg)`
✓	**Shoved front to back at the top: strength** Yield ÷ peak stress with 400 N pushing the top front edge, plus self weight.	`safety factor 11.81 (need 2), worst A (leg)`
✓	**Loaded lower shelf: strength** Yield ÷ peak stress with 100 kg spread over the shelf, plus self weight.	`safety factor 19.42 (need 2), worst C (long rail)`
✓	**Loaded lower shelf: deflection** Largest movement of the loaded members.	`0.3 mm (limit 5.8 mm)`
✓	**Does not tip over** Free-standing it would tip with 300 N sideways at 900 mm high, so it is fixed down.	`anchored (step 9, an M8 floor anchor through the base of every levelling foot); free-standing factor 1.13`
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	`9 steps`
✓	**Top bars: strength** Each bar spans 487 mm between cross rails: 100 kg on the middle of one span, or its share of the rated load.	`safety factor 2.83 (need 2), sags 0.8 mm under 100 kg`
✓	**Shelf bars: strength** Each shelf bar spans 660 mm between the lower rails, with 100 kg on its middle.	`safety factor 2.09 (need 2)`
✓	**Gaps between the top bars** At least the gap asked for, evened out so the outer bars sit flush with the front and back.	`14 bars, 10.8 mm gaps (asked 10 mm)`

✓	**Work height** Standing work: about wrist height for a vise, elbow height for fine work.	900 mm
✓	**Room above the shelf** At least 300 mm between the shelf and the top frame for boxes and cases.	600 mm clear

2 Materials

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
40 × 40 × 2 square tube	14	12.8 m	3 × 6.00 m	30 kg	$72
40 × 20 × 2 rectangular tube, laid flat	35	35.7 m	7 × 6.00 m	63 kg	$143

Plate and sheet

Mark	Part	Size	Qty	Step	Cost
P1	Foot plate	40 × 40 × 6 steel	4	7	$2.40

Parts to buy

Part	Spec	Qty	Step	Cost
Hex nut, M12	plain steel, welded to the foot plate	4	7	$0.60
Levelling foot, M12, bolt-down base	40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating	4	9	$16
Floor anchor, M8	sleeve or wedge anchor, to suit the levelling foot base hole	4	9	$4.80

Consumables

- MIG wire, 0.8 mm · 1 kg spool
- Shielding gas, Ar/CO2 · one small cylinder
- Cut-off discs, 1 mm · 5
- Flap discs, 80 grit · 3
- Primer and enamel · 1 L each
- Degreaser and rags

Tools

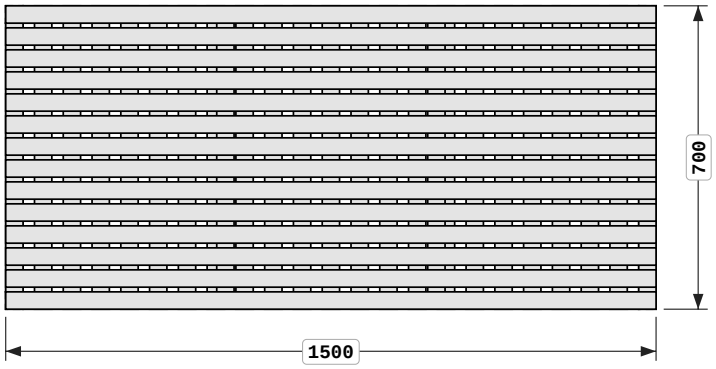
- MIG welder
- Hammer drill and masonry bit for the anchors
- Chop saw or angle grinder with cut-off discs
- Angle grinder with flap disc
- Welding table or flat floor
- Square, tape, scriber, spirit level
- Six or more clamps
- Drill, 13 mm bit, M12 tap
- A spacer block for the bar gaps

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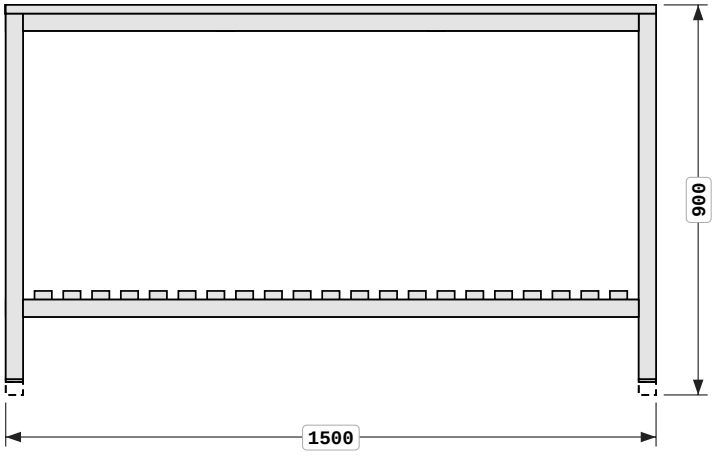
3 Drawings

Overall dimensions. Every number on this page is from the same model as the 3D view.

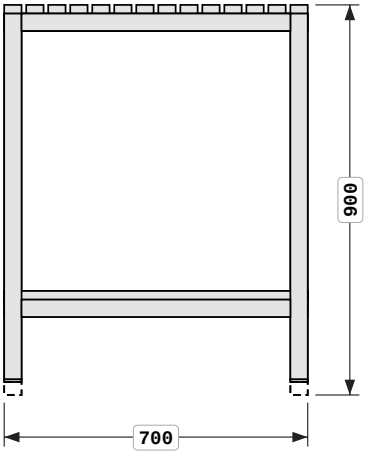
Top



Front



Side (from the right)



4 Legend

Units	millimetres, kilograms, degrees
Datum	x from the left end of the top, y from the front edge, z up from the floor
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-front, bottom-right…)
Joints	mitre = same section, on the bisector · butt onto X = square onto the face of X · flush with X = cut level with X's outer face · then weld
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm · tacked ±1 mm · finished ±2 mm · diagonals equal within 3

5 Order

Cut everything, tack the end frames, join them, weld out, weld the bars on, then feet, paint and load test.

Fabricate

1. Cut, deburr, foot plates 3 h
2. Two end frames (tack) 2 h
3. Join the end frames (tack) 2 h
4. Cross rails (tack) 1 h
5. Weld out the frame 3 h
6. Top and shelf bars 4 h
7. Foot plates 1 h

Assemble

9. Feet, anchors and load test 2 h

Finish

8. Clean and paint 2 h

6 Steps

Ticks save on this device. "Show in 3D" shows the build after that step, the new parts lit.

01 Cut, deburr, foot plates

Fabricate · 3 h

- [ ] Safety glasses, gloves and hearing protection from here on. Use bare or oiled tube, never galvanised.
- [ ] Cut every piece from the cutting plan, square ends; check each cut with a square.
- [ ] Deburr every end and write its piece mark on it in marker.
- [ ] Lay the pieces out by mark and count them against the cut list: 4 legs (A), 45 rails.
- [ ] Cut 4 foot plates 40 × 40 from 6 mm plate and drill a 13 mm hole in the centre of each.

Check Every piece within ±1 mm of its long point on the cut list.

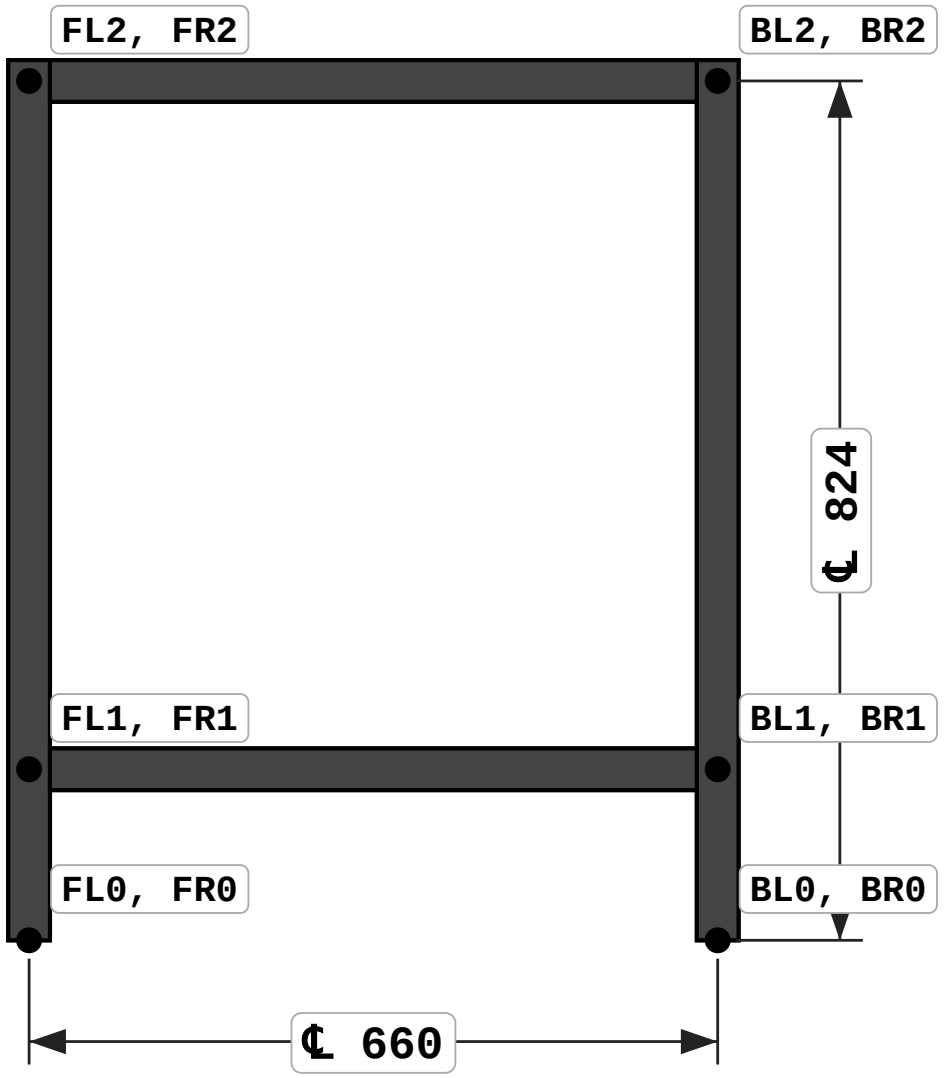
02 Two end frames (tack)

Fabricate · 2 h

- [ ] Lay two legs (A) flat on the welding table, parallel, outer faces against a straight edge.
- [ ] Fit a short rail (B) between them at the top, its top face flush with the leg tops, and a second one lower down with its centre line 164 up from the leg bottoms.
- [ ] Set the outside width to 700 at the top and the bottom, clamp each joint square, tack each joint on two faces.
- [ ] Build the second end frame on top of the first so the two match.

Check Diagonals of each frame equal within 2 mm; both frames the same within ±1 mm; flat on the table at all four corners.

Step 2: Side (from the right)



Joint positions

Point	x	y	z
FL0	20	20	36
FR0	1480	20	36
BL0	20	680	36
BR0	1480	680	36
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	860
FR2	1480	20	860
BL2	20	680	860
BR2	1480	680	860

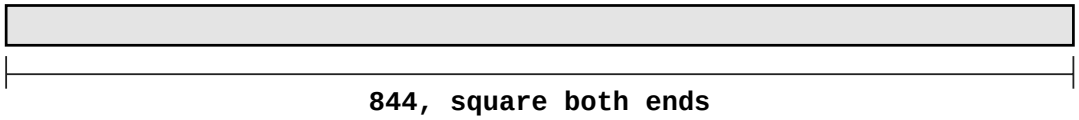
Pieces: A x4, B x4 (of 6)

A x4 — **844 mm**

Leg · SHS 40×40×2 · step 2

long point

`leg-FL, leg-FR, leg-BL, leg-BR`



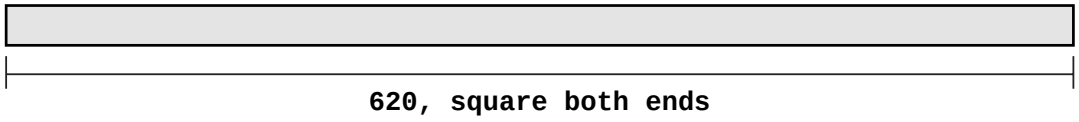
Left end: onto P1 (foot plate) · Right end: flush with B (short rail)

B x6 — **620 mm**

Short rail · SHS 40×40×2 · step 2

long point

`end-L-top, end-L-low, end-R-top, end-R-low, cross-top-1, cross-top-2`



Left end: butt onto A (leg) or C (long rail) · Right end: butt onto A (leg) or C (long rail)

03 Join the end frames (tack)

Fabricate · 2 h

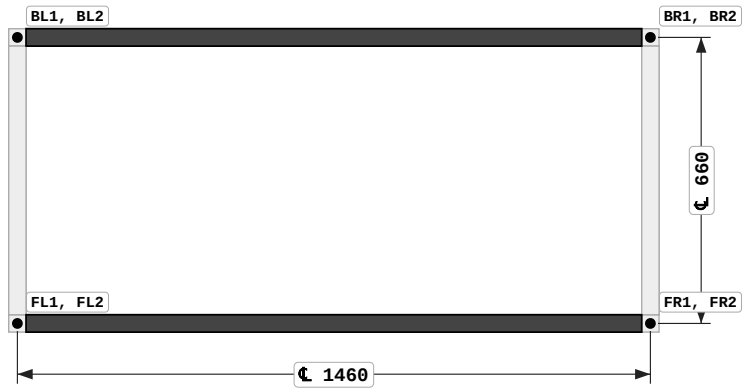
- [ ] Stand both end frames up on the table, top ends up, and brace each with a clamp to a square.
- [ ] Clamp the long rails (C) between the end frames: top rails flush with the leg tops, lower rails at the same height as the lower end rails.
- [ ] Set the outside length to 1500 on the top and the bottom rails.
- [ ] Tack the top long rails first, then the lower ones, one tack per joint.
- [ ] Measure the top diagonals, nudge the frame until they are equal, then tack every joint on two faces.

Check Top diagonals equal within 3 mm; every leg plumb within ±1 mm over its height; the frame sits on all 4 legs without rocking.

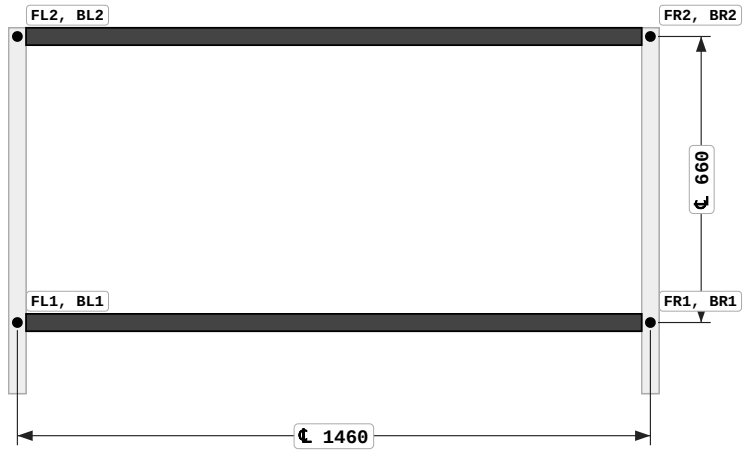
> Tack only: the frame is welded out in step 5, once every rail is in.

Jig

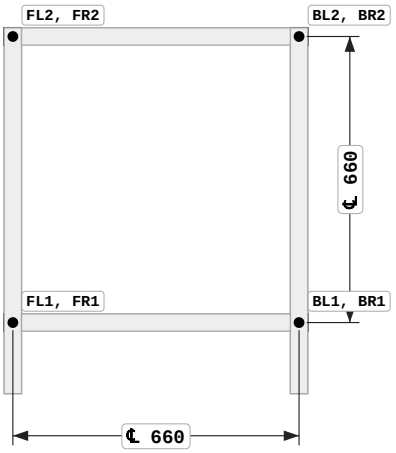
Step 3: Top



Step 3: Front



Step 3: Side (from the right)



Joint positions

Point	x	y	z
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	860
FR2	1480	20	860
BL2	20	680	860
BR2	1480	680	860

Pieces: C x4

C x4

1420 mm

Long rail · SHS 40×40×2 · step 3

long point

`long-F-top, long-B-top, long-F-low, long-B-low`



Left end: butt onto A (leg) · Right end: butt onto A (leg)

04 Cross rails (tack)

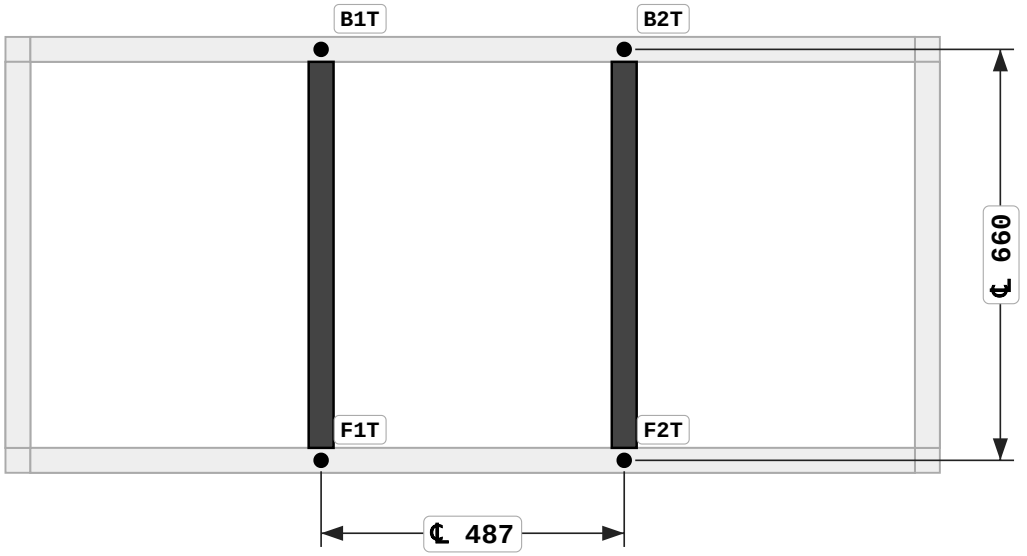
Fabricate · 1 h

- [ ] Mark the cross rail positions on the top long rails, front and back: 507, 993 from the left end to the centre line of each cross rail.
- [ ] Fit a short rail (B) as a cross rail at each mark, between the top long rails, top face flush with them, square to the front rail, and tack both ends.

Check Cross rails flush with the top within ±1 mm (lay a straight edge across), and 487 apart centre to centre within ±2 mm.

Jig

Step 4: Top



Joint positions

Point	x	y	z
F1T	507	20	860
F2T	993	20	860
B1T	507	680	860
B2T	993	680	860

Pieces: B x2 (of 6)

B: cut sheet in step 2

05 Weld out the frame

Fabricate · 3 h

- [ ] Re-check the top diagonals and the leg plumb before any weld; fix them while it is only tacked.
- [ ] Weld every joint all round in short runs, alternating diagonally opposite corners so the heat does not pull the frame out of square.
- [ ] Weld the top frame joints first, then the lower frame, then the cross rails. Leave it clamped until cold.
- [ ] Grind the welds on the top faces of the top frame and the lower frame flush, so the bars sit flat.

Check Top diagonals still equal within 3 mm; a straight edge across the rail tops shows no gap over 1 mm.

> Ventilate: weld fumes from mill scale and paint are toxic.

06 Top and shelf bars

Fabricate · 4 h

- [ ] Lay the 14 top bars (D) flat on the top frame, running the length, ends flush with the frame ends.
- [ ] Put the outer two flush with the front and back of the frame and space the rest evenly: 10.8 mm gaps (cut a spacer block to that width and use it at both ends of every bar).
- [ ] Tack each bar to the end rails at both ends, checking the gaps as you go, then tack it to every cross rail.
- [ ] Lay the 21 shelf bars (E) flat across the lower long rails, front to back, ends flush with the front and back of the frame, between the legs with 26 mm gaps (the same gap next to each leg). Tack both ends of each.
- [ ] Stitch-weld each bar to every rail it crosses, 25 mm on each side, working from the middle bar outward (98 crossings).
- [ ] Grind any weld that stands proud of a bar top flush.

Bars

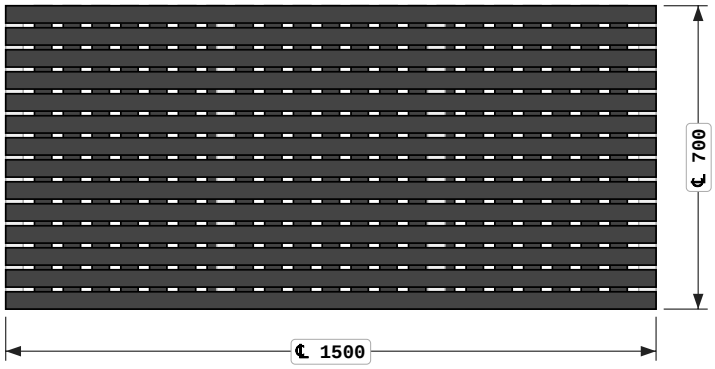
Where	Bars	Runs	Gap	First centre line
Top	D x14	left to right	10.8	20 from the front
Shelf	E x21	front to back	26	86 from the left end

Check Gaps equal within 1 mm; a straight edge laid across the bars rocks nowhere; no bar moves when you lever it with a bar.

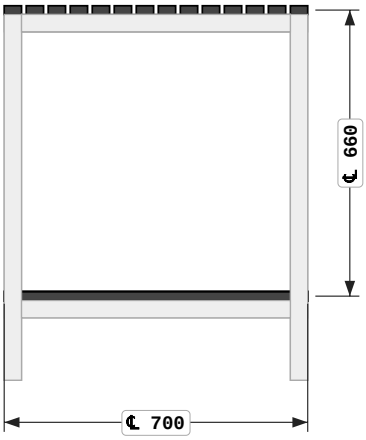
> Weld the crossings in an even pattern, never one bar end to end: the top stays flat.

Jig

Step 6: Top



Step 6: Side (from the right)

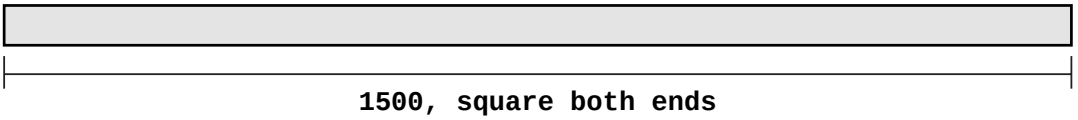


Pieces: D x14, E x21

D x14 — 1500 mm

Bar · RHS 40×20×2 · step 6 — long point

`top-bar-1, top-bar-2, top-bar-3, top-bar-4, top-bar-5, top-bar-6, top-bar-7, top-bar-8, …and 6 more`



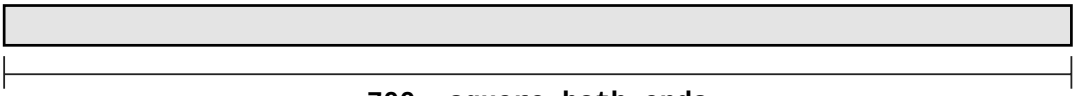
1500, square both ends

Left end: free end · Right end: free end

E x21 — 700 mm

Shelf bar · RHS 40×20×2 · step 6 — long point

`shelf-bar-1, shelf-bar-2, shelf-bar-3, shelf-bar-4, shelf-bar-5, shelf-bar-6, …and 15 more`



700, square both ends

Left end: free end · Right end: free end

07 Foot plates

Fabricate · 1 h

- [ ] Weld an M12 nut on the top face of each foot plate, centred over the 13 mm hole: thread a bolt through both to hold it straight while you weld.
- [ ] Lay the bench on its back (two people). Centre a foot plate on the bottom of each leg, nut inside the tube, and weld it all round.
- [ ] Chase each nut with an M12 tap or run a bolt in and out to clear spatter.

Check A bolt screws into every nut by hand, straight.

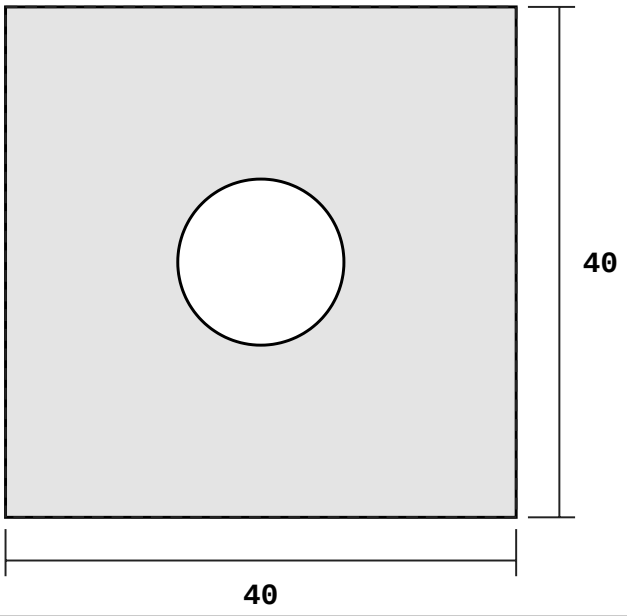
Parts

- Hex nut, M12 ×4 plain steel, welded to the foot plate

Plates

P1 x4

Foot plate · 6 mm steel · step 7



1 hole ø13 mm, centres from the lower-left corner: (20, 20).

Dashed: where a member meets its face (welded on, or under it), from the lower-left corner: (0, 0).

Weld an M12 nut over the hole on the top face first, then weld the plate centred under the leg, all round.

08 Clean and paint

Finish · 2 h

- [ ] Degrease, wire-brush the welds, and sand any mill scale on show.
- [ ] Prime and paint with two coats of enamel, getting the paint into the gaps between the bars; let it dry overnight.
- [ ] Leave the bar tops bare and oiled instead if you want a metal working surface.

Check No bare steel left except the oiled bar tops; paint dry to the touch.

09 Feet, anchors and load test

Assemble · 2 h

- [ ] Screw a levelling foot into each leg and set each to 30 under the foot plate.
- [ ] Stand it in place and level it with the feet: a level on the top, front to back and end to end. Lock each foot with its nut.
- [ ] Mark the anchor hole of every foot base on the floor, move the bench, drill for the M8 anchors, set them, and bolt every foot down (4 anchors).
- [ ] Load it gradually, in steps of about a quarter, up to 300 kg spread over the top, and leave it an hour.

Check Level within ±1 mm over the length; does not move or lift when pushed hard at the top edge; no visible sag or cracked weld after the load test.

Parts

- Levelling foot, M12, bolt-down base x4 40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating
- Floor anchor, M8 ×4 sleeve or wedge anchor, to suit the levelling foot base hole

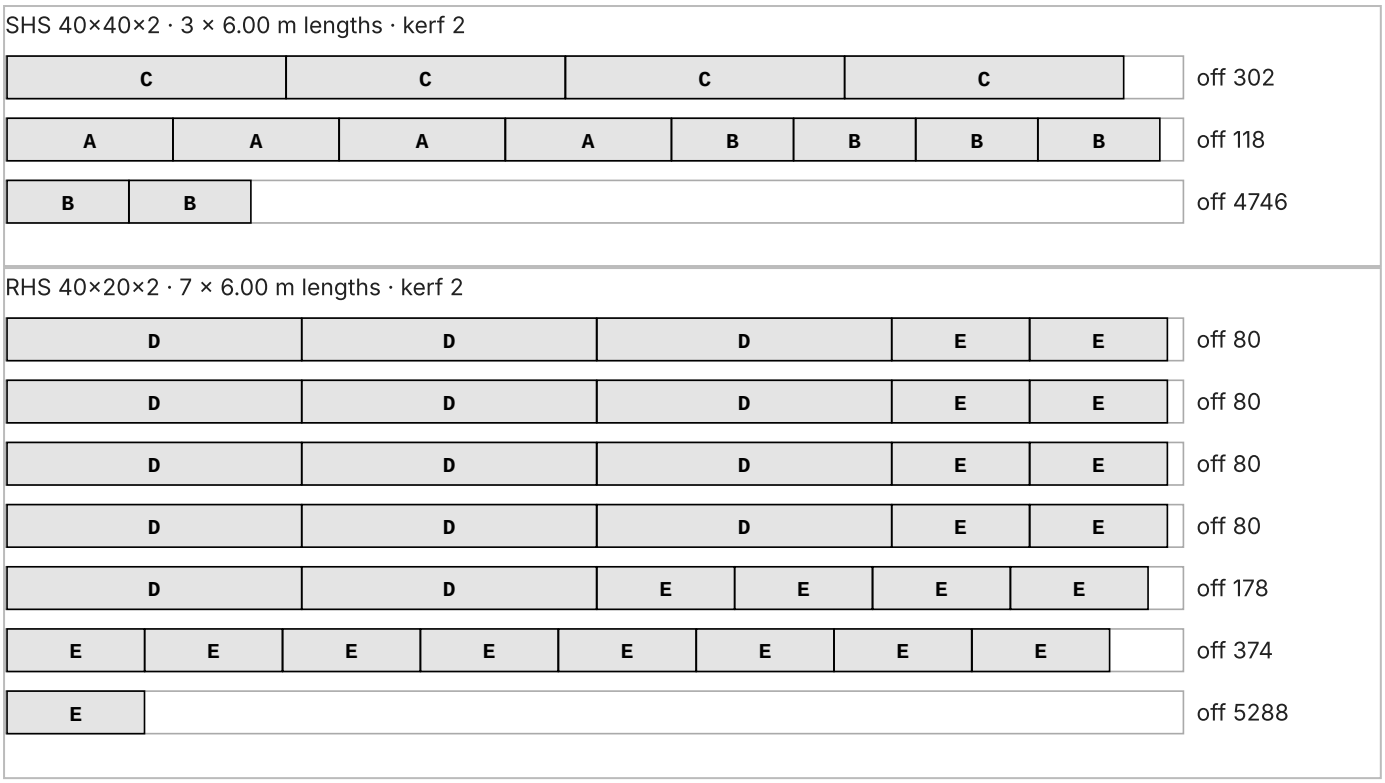
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7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Leg	4	SHS 40×40×2	844	square · onto P1 (foot plate)	square · flush with B (short rail)	2
B Short rail	6	SHS 40×40×2	620	square · butt onto A (leg) or C (long rail)	square · butt onto A (leg) or C (long rail)	2
C Long rail	4	SHS 40×40×2	1420	square · butt onto A (leg)	square · butt onto A (leg)	3
D Bar	14	RHS 40×20×2	1500	square · free end	square · free end	6
E Shelf bar	21	RHS 40×20×2	700	square · free end	square · free end	6

Cutting plan

Each bar is one stock length; cut the pieces left to right.



Welded steel workbench · v1. Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.